EE 210L: Digital Circuits and Logic Design Lab

Lab 7: Multiplexers and Demultiplexers

Name: _____ Date: _____

Partner(s): _____ Section: _____

Record logic levels, 0 or 1, and the measured voltage only where a column asks for it. Read levels directly: near 0 V is a 0, near 3.4 V is a 1.

Part 5.1: Basic Multiplexer Operation

Data inputs wired: D0, D2, D4, D6 = +5 V D1, D3, D5, D7 = GND

S2	S1	S0	Selected input	Y	W
0	0	0			
0	0	1			
1	0	0			
1	0	1			

Is W always the complement of Y?

_____

Part 5.2: The Enable Input

Select code used: _____

Y with $\bar{E} = 0$: _____ V Y with $\bar{E} = 1$: _____ V

What does the enable input do, and why would a shared data bus need one?

_____

_____

Part 5.3: A 3-Variable Function on the MUX

$F(A,B,C) = A'BC + AB'C' + AB$

Fill in which data inputs you tied HIGH, then record measurements.

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A (S2)	B (S1)	C (S0)	D_n tied to	Measured V_{out} (V)	Logic Output	Expected F
0	0	0				
0	0	1				
0	1	0				
0	1	1				
1	0	0				
1	0	1				
1	1	0				
1	1	1				

How many logic gates did this function require?

Part 5.4: Decoder as a Demultiplexer, with LEDs

Square wave into pin 6: frequency _____ Hz amplitude _____ V

LEDs on Y2, Y5 and Y7 only. For each address, record which LED blinks and what the other two do.

A2	A1	A0	Expected output	Which LED blinks	What the other two do
0	1	0	Y2		
1	0	1	Y5		
1	1	1	Y7		

Does the blinking LED match the address you set? Explain the relationship.

Discussion and Conclusions

Address at least: why implementing a function on a MUX needs no minimization, what the enable input is for, and how a decoder becomes a demultiplexer.

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